A Reverse-Engineered College Basketball Rating Model

A KenPom-Inspired Algorithm

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Executive summary. This report documents a college basketball rating system that I built to approximate the predictive logic behind Ken Pomeroy's KenPom system while still remaining fully tunable. The model converts box score data into possession-based efficiencies, caps blowout margins, adjusts for home-court advantage, rescales performances by opponent strength, and weights recent games more heavily than older ones. The resulting rankings are directionally similar to KenPom's board while still expressing a few meaningful differences. Those differences are exactly the point: by building my own system, I can inspect every assumption, test alternatives, and understand why the model likes or dislikes a team. The model itself is unchanged from the original write-up; the updates in this version are factual and interpretive, reflecting the completed 2026 NCAA tournament.

Introduction / Motivation

In the NCAA, there are 364 Division I schools, all of which belong to a conference of roughly 8–18 schools. Each school plays about 30–35 games per season, around 10–15 of which are against teams outside of its conference. At the end of each season, the NCAA men's basketball tournament begins with 68 teams and ultimately narrows to a 64-team main bracket after the First Four, producing a single national champion. Because the field is so large and cross-conference scheduling is limited, there are many tournament matchups between teams that never faced one another during the regular season.

It is because of this that analytical metrics like KenPom exist. KenPom is a system created by Ken Pomeroy that aims to quantify how well each team has performed while adjusting for quality of opponents. KenPom is useful because, by adjusting for strength of schedule, it lets us compare teams that have never played each other and may share few common opponents. However, Pomeroy keeps his exact algorithm private, so I decided to build my own.

I wanted to build my own version of a KenPom-style rating system for two reasons. First, I wanted a predictive metric that reflects the way modern basketball analysts think about team strength: not just wins and losses, but efficiency, pace, opponent quality, game location, and recent form. Second, I wanted a model that I could tweak myself. Public systems such as KenPom are extremely useful, but the exact formula is proprietary. Rebuilding a similar framework from scratch forces me to think carefully about what each modeling choice is supposed to measure.

KenPom has described his own ratings as a *predictive* system meant to estimate how strong a team would be "if it played tonight." He also emphasizes adjusted efficiency margin—offensive efficiency minus defensive efficiency—as a clean measure of team strength over 100 possessions.[1] That is the exact spirit of this project. My goal is not to duplicate the private formula perfectly. My goal is to

[1] Ken Pomeroy, "Ratings Explanation," KenPom blog, Nov. 29, 2006; and "Ratings Methodology Update," Oct. 4, 2016.

build a transparent approximation that behaves like a serious predictive model and can be improved over time.

The Algorithm

Assumptions

When designing this system, I made several basketball-logic assumptions that are reflected in the algorithm. The overarching principle can be broken down into two parts:

- Binary wins and losses matter much less than general performance. A close win versus a really bad team is much worse than a close loss versus a really good team.

Although that may be the most important assumption, it is not the only one:

- Possession-based efficiencies are much more useful than raw points scored or allowed. A team can easily score more points if its game had many more possessions, but in a theoretical head-to-head matchup, both teams will have the same number of possessions with and without the ball.
- Blowouts are not infinitely informative. For example, outscoring a team by 40 points is treated the same as only winning by 30.
- Home-court advantage affects both offensive and defensive efficiencies. For example, a team that had an offensive efficiency of 90 at home did worse than a team with the same offensive efficiency on the road.
- Lastly, recently played games are more significant than games played a while ago.

What my metric is trying to measure

At a high level, the metric estimates how many points per 100 possessions a team would score and allow against the average Division I opponent on a neutral floor. The final rating is:

$$\text{Adjusted Net Rating} = \text{Adjusted Offensive Efficiency} - \text{Adjusted Defensive Efficiency}.$$

This mirrors KenPom's adjusted efficiency margin framework. In his system, AdjEM is the number of points a team would be expected to outscore an average Division I opponent by over 100 possessions.[2] My model uses the same target quantity because it is both predictive and interpretable.

Data collection pipeline

The raw dataset was built by scraping the ncaa.com website, which allowed me to map each game to its calendar date from the daily scoreboard feed, and then pull team-level box score statistics for every completed game. For each team-game observation, the scraper stores the fields needed for an efficiency model: shooting volume, offensive rebounds, turnovers, free throws, made threes, made field goals, and made free throws.

[2] Ken Pomeroy, "Ratings Methodology Update," KenPom blog, Oct. 4, 2016.

This is an important design choice. The model is based entirely on reproducible box score data rather than subjective information. That makes the pipeline auditable end to end: from API call, to raw CSV, to adjusted ratings.

Possessions and raw scoring output

The algorithm starts from a standard possession estimate:

$$\text{Possessions} = \text{FGA} - \text{ORB} + \text{TO} + 0.475 \times \text{FTA}.$$

This is a common college basketball approximation and is consistent with possession-based efficiency analysis. Points scored are reconstructed from the box score as:

$$\text{Points Scored} = 3 \times \text{3PM} + 2 \times (\text{FGM} - \text{3PM}) + \text{FTM}.$$

The purpose of these two steps is straightforward: remove pace from the evaluation. A team should not look elite merely because it plays quickly. By converting performance to a per-100-possession basis, the metric compares teams on the quality of their possessions rather than the quantity of them.

Margin capping

One of the most important nuances in the formula is margin capping. Rather than using the actual score margin directly, the algorithm clips it to a symmetric threshold:

$$\text{Margin}_{\text{capped}} = \min\{C, \max[-C, \text{Margin}]\},$$

where C is a tunable margin cap. After tuning, the best value in this project was $C = 12$.

Why cap margin at all? Because predictive models should reward dominance, but they should not let a handful of extreme blowouts overpower the signal from the rest of the schedule. KenPom has written that scoring margin does contain predictive information, including at larger margins, but that the relationship is not simple and must be handled carefully.[3] My compromise is to preserve the direction and general strength of a result while limiting the ability of a couple runaway games to distort the season-long profile.

Once the margin is capped, the total points in the game are redistributed into capped scored and allowed values:

$$\text{Points Scored}_{\text{capped}} = \frac{\text{Total Points} + \text{Margin}_{\text{capped}}}{2},$$
$$\text{Points Allowed}_{\text{capped}} = \frac{\text{Total Points} - \text{Margin}_{\text{capped}}}{2}.$$

Those values are then converted into offensive and defensive efficiencies:

[3] Ken Pomeroy, "Evidence that scoring margin matters," KenPom blog, Jan. 22, 2013.

$$\text{Offensive Efficiency} = 100 \times \frac{\text{Points Scored}_{\text{capped}}}{\text{Possessions}},$$
$$\text{Defensive Efficiency} = 100 \times \frac{\text{Points Allowed}_{\text{capped}}}{\text{Possessions}}.$$

This is a strong modeling choice because it treats a 30-point win as clearly better than a 5-point win, but not six times as informative.

Home-court adjustment

After the initial efficiency calculation, the notebook applies a fixed home-court adjustment by shifting the home team's offense and defense in opposite directions, and doing the reverse for the road team. The tuned home-court value is $H = 2.0$ points per 100 possessions:

$$\text{Home Offensive Efficiency} \leftarrow \text{Home Offensive Efficiency} - H,$$
$$\text{Home Defensive Efficiency} \leftarrow \text{Home Defensive Efficiency} + H,$$
$$\text{Away Offensive Efficiency} \leftarrow \text{Away Offensive Efficiency} + H,$$
$$\text{Away Defensive Efficiency} \leftarrow \text{Away Defensive Efficiency} - H.$$

The reasoning is that location affects observed efficiency. If a home team posts a certain raw efficiency, part of that performance should be attributed to venue advantage rather than pure team quality. KenPom has written extensively about home-court advantage and the difficulty of estimating it precisely, noting that it is real but noisy and generally smaller than many fans assume.[4] My model follows that same logic with a simpler constant adjustment.

Opponent adjustments

Raw efficiency is still incomplete because it ignores who the team played. Scoring 112 points per 100 possessions against an elite defense is more impressive than scoring the same number against a weak one. To fix that, the model rescales each game's efficiency by the opponent's current estimated strength relative to the national average efficiency:

$$\text{AdjOff}_g(A) = \text{OffEff}_g(A) \times \frac{\text{National Avg}}{\text{Opp Current AdjDef}},$$
$$\text{AdjDef}_g(A) = \text{DefEff}_g(A) \times \frac{\text{National Avg}}{\text{Opp Current AdjOff}}.$$

This is the heart of the algorithm. It creates an iterative, opponent-aware rating process. Strong performances against strong competition get amplified. Strong-looking box score lines against weak teams get pulled back toward the mean.

Two implementation details are especially important here:

[4] Ken Pomeroy, "Mining point spread data: home court advantage," KenPom blog, May 21, 2018.

- If a team has no prior history, the model falls back to the national average efficiency. This prevents the early-season ratings from exploding due to missing priors.
- For a team's first game, the notebook keeps the raw efficiency rather than forcing a questionable opponent adjustment off almost no information. This is a conservative choice that improves stability.

Recency weighting

The final major nuance is recency weighting. Instead of taking a plain average of all adjusted efficiencies, the notebook uses an exponentially decaying weighted average. If a team's kth most recent game is indexed by k, their weights are:

$$w_k = 0.5^{k/\text{half-life}}.$$

The best half-life found by tuning was 20 games. Intuitively, that means a game from roughly 20 contests ago receives about half the weight of the most recent one.

This is one of the clearest places where I intentionally leaned into the predictive framing. A team in March is not the same team it was in November. Rotations tighten, freshmen develop, injured players return, and coaching adjustments accumulate. KenPom's public explanations also frame the ratings as a "play tonight" system rather than a full-season achievement score.[5] My recency weighting is a direct attempt to encode that idea.

Hyperparameter tuning

To avoid choosing the key knobs by feel alone, I evaluated combinations of the margin cap, home-court advantage, and half-life against out-of-sample style game predictions. For games from Jan. 1, 2026 onward (to avoid early-season sample variation), I formed a pregame rating for each team and predicted home margin as:

$$\widehat{\text{Home Margin}} = \text{Home AdjNet}_{\text{pre}} - \text{Away AdjNet}_{\text{pre}} + H.$$

The score used for tuning is mean absolute error between predicted and actual home margin. The best parameter set found in the notebook is shown below.

Configuration	Margin cap	Home-court adj.	Half-life
Initial trial	16	3.75	12
Best tuned model	12	2.0	20

Table 1: Model tuning results. The final report uses the tuned configuration.

This tuning step is important because it turns the project from a hand-built formula into a tested predictive model. It also gives every modeling choice a measurable consequence.

[5] Ken Pomeroy, "Ratings Explanation," KenPom blog, Nov. 29, 2006.

Results

Top of the board

Using the tuned settings $(C, H, \text{half-life}) = (12, 2, 20)$, the model's top 15 teams—measured on the original pre–Sweet 16 snapshot through games of Sunday, March 22, 2026—were as follows:

Rank	Team	Adj. Off.	Adj. Def.	Adj. Net
1	Duke	121.36	97.53	23.83
2	Arizona	120.50	96.84	23.66
3	Michigan	119.54	96.72	22.82
4	Purdue	126.25	104.97	21.28
5	Houston	118.23	97.42	20.80
6	St. John's (NY)	115.39	95.48	19.91
7	Illinois	123.66	103.99	19.67
8	Gonzaga	116.74	97.32	19.42
9	Saint Mary's (CA)	120.16	100.78	19.38
10	UConn	117.29	97.91	19.38
11	Iowa St.	117.13	97.98	19.15
12	Virginia	118.19	99.33	18.87
13	Florida	118.63	99.95	18.68
14	Michigan St.	118.06	99.89	18.17
15	Texas Tech	121.05	103.40	17.65

Table 2: Top 15 teams in the custom rating model, using the March 22 snapshot.

At the top, the model passes a basic sanity check: the same general group of contenders appears near the top as on KenPom's board. As of games through Sunday, March 22, 2026, KenPom's top 10 were Michigan, Arizona, Duke, Houston, Florida, Illinois, Iowa State, Purdue, Michigan State, and Connecticut.[6] My model also places Duke, Arizona, Michigan, Purdue, Houston, Illinois, Iowa State, Michigan State, and UConn in the upper tier, which is a strong sign that the architecture is capturing real signal rather than producing arbitrary outputs.

To make that comparison more concrete, 18 of my top 25 teams also appeared in KenPom's top 25, and the Spearman rank-order correlation coefficient on those shared teams is approximately 0.71. That is not a perfect match, but it is exactly what I would expect from an independent model: broad overall agreement with a few meaningful deviations driven by different assumptions.

Where my model differs

The most interesting part of the project is not where the model agrees with KenPom. It is where it disagrees and gives me something to investigate. A recency-weighted framework will naturally diverge from a benchmark whenever it decides a team's current form matters more than its larger season profile.

Those gaps are useful because they reveal what the formula is emphasizing. A higher St. John's placement suggests my model bought into the Red Storm's late-season form more aggressively than KenPom did. A higher Saint Mary's placement suggests my model was more willing to reward the Gaels' efficiency profile than the benchmark was. A lower Florida placement suggests my model was

[6] KenPom ratings page, data through games of Sunday, March 22, 2026.

less persuaded by the Gators' full-season dominance once margin capping and recency weighting were applied.

Post-tournament read-through

Now that the 2026 NCAA tournament is complete, this pre–Sweet 16 snapshot can be evaluated against the full bracket rather than against just the first weekend. Michigan won the national championship on April 6, defeating UConn 69–63 in Indianapolis for the program's first national title since 1989.[7]

That outcome cuts both ways for the model in an interesting way. On one hand, Michigan was ranked third in my system and UConn 10th, so the championship game was still populated by teams my model viewed as genuine contenders. On the other hand, the model's No. 1 team, Duke, did not reach the Final Four, losing 73–72 to UConn in the Elite Eight on a last-second three.[8] That is not an indictment of the model by itself—single-elimination tournaments are inherently high variance—but it is a reminder that even a strong predictive ranking is not the same thing as a bracket oracle.

Several specific cases are more revealing than the title game itself:

- **Michigan looks like a real success for the model.** My ratings had Michigan third before the Sweet 16, which in hindsight looks quite strong given that the Wolverines finished the job and won the national title.
- **UConn validates the broader top tier.** My model had UConn 10th before the Sweet 16, and the Huskies went on to reach the national championship game.
- **Saint Mary's was a clear miss.** My model ranked Saint Mary's ninth nationally, while KenPom had the Gaels much lower. Saint Mary's then lost 63–50 to Texas A&M in the first round.[9]
- **Florida is the strongest argument for the model's independence.** My system ranked Florida 13th, while KenPom had Florida fifth on March 22. Florida then lost 73–72 to Iowa in the second round.[10] One upset does not prove my model was "right," but it does show the value of an independent framework that can be more cautious than a respected benchmark.
- **St. John's was not as far-fetched as it first looked.** My model ranked the Red Storm sixth, well above KenPom's placement at the time, and St. John's advanced to the Sweet 16 before losing 80–75 to Duke.[11]

The bigger lesson is that the model produced a mix of strong calls and obvious misses, which is exactly what an honest evaluation should show. A useful predictive system should not be judged by

[7] NCAA.com, "Michigan wins the 2026 DI men's basketball championship," Apr. 6, 2026; and NCAA championship history page.

[8] NCAA.com tournament schedule and results page; ESPN recap of UConn vs. Duke, Mar. 29, 2026.

[9] ESPN game summary, Texas A&M 63, Saint Mary's 50, Mar. 19, 2026.

[10] NCAA.com video recap of Iowa vs. Florida, Mar. 22, 2026; Florida Gators official recap, Mar. 22, 2026.

[11] NCAA.com condensed game recap, Duke vs. St. John's, Mar. 27, 2026.

whether every tournament upset matches the board. It should be judged by whether its inputs are sensible, whether its overall ordering is plausible, and whether its disagreements are interpretable.

Selected disagreements between my model and KenPom

My rk	Team	KenPom rk(Mar. 22)	Diff.	Post-tournament interpretation
3	Michigan	1	-2	Slightly lower than KenPom, but still firmly in title-contender range; Michigan went on to win the national championship.
6	St. John's (NY)	16	+10	My model was much more bullish on the Red Storm's late form; they advanced to the Sweet 16 before losing a competitive game to Duke.
9	Saint Mary's (CA)	26	+17	This was one of the clearest misses in the model; the Gaels lost 63–50 to Texas A&M in the first round.
13	Florida	5	-8	My system was materially lower on Florida than KenPom was, and Florida was then upset by Iowa in the second round.

Table 3: Selected ranking disagreements between my model and KenPom, with the benefit of completed tournament results.

What these results say about my approach

Overall, the results show three things.

First, the model is **basketball-native**. It is built from possession and efficiency logic, site adjustment, opponent quality, and recency rather than generic standings data.

Second, the model is **predictively oriented**. I tuned it against margin prediction error, not against the AP poll, NCAA seeds, or subjective reputation.

Third, the model is **interpretable**. When it disagrees with KenPom, I can usually explain why. That matters in analytics settings because decision-makers need more than a ranking; they need a rationale.

The completed tournament also adds a fourth takeaway: the model appears good enough to identify the true contender pool, but not immune to overreacting in a few recency-driven cases. That is probably the most realistic conclusion possible. Predictive ratings should help narrow the set of likely outcomes and spotlight mispriced teams, not pretend randomness has disappeared.

Conclusion

Building this metric was an exercise in both sports knowledge and modeling discipline. The final system is not presented as the secret KenPom formula. Instead, it is my best transparent approximation of the same predictive philosophy. It transforms raw NCAA box scores into a tuned, opponent-adjusted, recency-aware rating that behaves plausibly, tracks a respected benchmark, and produces testable independent views.

With the full 2026 NCAA tournament now complete, the model's pre–Sweet 16 snapshot looks

neither perfect nor arbitrary. It had Michigan and UConn comfortably inside its top 10 before they met in the national championship game, and it was notably more skeptical of Florida than KenPom was before the Gators' second-round exit. At the same time, it clearly overrated Saint Mary's. That combination of hits and misses is useful. It means the system is independent enough to generate original opinions, but structured enough that those opinions can be evaluated and improved.

References

1. Ken Pomeroy, “Ratings Explanation,” KenPom blog, Nov. 29, 2006.
2. Ken Pomeroy, “Ratings Methodology Update,” KenPom blog, Oct. 4, 2016.
3. Ken Pomeroy, “Evidence that scoring margin matters,” KenPom blog, Jan. 22, 2013.
4. Ken Pomeroy, “Mining point spread data: home court advantage,” KenPom blog, May 21, 2018.
5. KenPom ratings page, data through games of Sunday, March 22, 2026.
6. KenPom ratings page, data through games of Monday, April 6, 2026.
7. NCAA.com, “Michigan wins the 2026 DI men’s basketball championship,” Apr. 6, 2026.
8. NCAA.com, “2026 March Madness: Men’s NCAA tournament schedule, dates,” updated Apr. 6, 2026.
9. NCAA.com history page, “DI Men’s Basketball Championship History,” accessed Apr. 17, 2026.
10. ESPN, “Texas A&M 63–50 Saint Mary’s,” game summary, Mar. 19, 2026.
11. NCAA.com video recap, “Defending champs out: Iowa stuns No. 1 Florida in March Madness second round,” Mar. 22, 2026.
12. Florida Gators official recap, “FINAL: Iowa 73, Florida 72,” Mar. 22, 2026.
13. NCAA.com condensed game recap, “Duke vs. St. John’s,” Mar. 27, 2026.
14. ESPN, “UConn rallies, stuns Duke on last-second 3 to reach Final Four,” Mar. 29, 2026.